# Dilmurod Valiev

AI/ML researcher — neuro-symbolic AI, 3D digital twins, speech interfaces

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PhD candidate building speech interfaces for 3D digital twins: a language model turns an instruction into a symbolic command, and a deterministic validator either executes it on the twin or refuses it with a reason. Production web and mobile work in React, Laravel and Flutter alongside the research. Looking for digital twin, industrial simulation or applied ML engineering roles.

## EXPERIENCE

**AI/ML researcher** — Inha University AI Lab, Incheon  Mar 2021 – present

- Built the speech-to-symbolic pipeline for aircraft-maintenance training on HoloLens: ASR, intent parsing and a symbolic command executed on a Unity digital twin.
- Designed the **action sandbox**, a deterministic validator that admits or refuses each command against twin state, procedure and user intent. Removing it raises unsafe actions from 0.7% to 11.8%.
- Extended speech recognition to Korean for the maintenance domain and benchmarked ASR and LLM back-ends for sequence-to-symbol generation.

**Full-stack developer** — Designmodo, Brooklyn, NY (remote)  Sep 2021 – Jan 2022

- Cut render time **52%** on multi-component previews in a React editor.
- Built a SlateJS text editor from scratch, with independent undo/redo history for editing and for document actions.
- Migrated legacy HTML projects to JSON through Laravel queues, with WebSocket progress reporting; reduced image storage by 24% with versioned backups.

**Flutter developer** — Findme, Togliatti, Russia (remote)  Aug 2021 – Jan 2022

- Shipped three consumer apps; improved rendering **27%** by narrowing widget rebuild scope.

## PROJECTS

**voice-to-twin-gate** — a spoken command becomes a checked action on a 3D twin  2026

`live demo · github.com/Dilmurod777/voice-to-twin-gate`

- Local LLM turns speech into a symbolic action; a deterministic gate checks twin state, procedure and 3D hierarchy, and executes or refuses it with a reason. **0 of 13** unsafe commands executed, against 13 of 13 with the LLM alone. FastAPI, Ollama, Three.js, 44 tests.

**manual-rag-evals** — three retrievers, one manual, 60 checked questions  2026

`github.com/Dilmurod777/manual-rag-evals`

- Vector, hybrid and reranked retrieval answer the same 60 questions over FAA AC 43.13-1B; **60/60** gold pages found, 55/60 answers cite the right page, and a distance gate refuses 9 of 20 unanswerable questions before generation. FastAPI, Ollama, NumPy, 17 tests in CI.

**llm-serving-bench** — Ollama vs llama.cpp vs vLLM on one GPU  2026

`github.com/Dilmurod777/llm-serving-bench`

- Same 8B model, 200 fixed prompts, one 24 GB card: **1646 tok/s** at 32 concurrent requests on vLLM at $0.06 per 1M output tokens, against 747 tok/s on Ollama at $0.13. `make bench`, 24 tests in CI.

## PUBLICATIONS

[1] Siyaev, A.; **Valiev, D.**; Jo, G.-S. Interaction with Industrial Digital Twin Using Neuro-Symbolic Reasoning. *Sensors* 2023, 23(3), 1729. `doi:10.3390/s23031729`

## EDUCATION

**Integrated MS/PhD, Inha University** — Incheon. CGPA 4.20/4.5  Mar 2021 – present

**BSc, Inha University in Tashkent**. First of 270 students, 4.48/4.5, honors, full scholarship.  2016 – 2020

## SKILLS

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|---|---|
| Languages | Python, TypeScript / JavaScript, C#, SQL |
| ML | PyTorch, TensorFlow / Keras, Whisper, LLM fine-tuning, structured output, RAG evaluation |
| 3D / XR | Unity, HoloLens / MRTK, Three.js, WebXR |
| Web | React, Next.js, Laravel, Flutter, FastAPI, Docker, GitHub Actions, PostgreSQL, MySQL, NoSQL |